

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract	Tut	Total
VSEC102	C Programming	--	2*+2	-		2	-	2

Course Code	Course Name	Examination Scheme							
		Theory					Term work	Pract / Oral	Total
		Internal Assessment Test (IAT)			End Sem Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	IAT-I + IAT-II (Total)					
VSEC102	C Programming	-	-	-	-	-	25	25	50

Lab Objectives: This subject aims to provide students with an understanding of the role computation can play in solving problems. The Course will be taught using C-Programming Language.

1. Understand and use basic terminology in computer programming.
2. Use various data types in C programs effectively.
3. Design and implement programs involving decision structures, loops, and functions.
4. Design Implement Arrays , String, and Structure
5. Describe and utilize memory dynamics through the use of pointers.
6. Use different data structures and create/update basic data files in C.

Lab Outcomes: Learners will be able to

1. Illustrate the basic terminology used in computer programming.
2. Use different data types in a computer program.
3. Design programs involving decision structures, loops and functions.
4. Implement Arrays , String, and Structure
5. Describe the dynamics of memory by the use of pointers.
6. Use different data structures and create/update basic data files.

DETAILED SYLLABUS

Sr. No	Module	Detailed Content	Hours	LO mapping
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1	Fundamentals of C-Programming	1.1 Character Set, Identifiers and keywords, Data types, Constants, Variables. 1.2 Operators-Arithmetic, Relational and logical, Assignment, Unary, Conditional, Bitwise, Comma, other operators. Expression, statements, Library Functions, Preprocessor. 1.3 Data Input and Output – getchar(), putchar(), scanf(), printf(), gets(), puts(), Structure of C program .	06	LO1, LO2
2	Control Structures	2.1 Branching - If statement, If-else Statement, Multiway decision. 2.2 Looping – while, do-while, for 2.3 Nested control structure- Switch statement, Continue statement Break statement, Goto statement.	05	LO3
3	Functions and Parameter	3.1 Function -Introduction of Function, Function Main, defining a Function, accessing a Function, Function Prototype, Passing Arguments to a Function, Recursion. 3.2 Storage Classes –Auto , Extern , Static, Register	05	LO3
4	Arrays , String Structure	4.1 Array-Concepts, Declaration, Definition, Accessing array element, One-dimensional and Multidimensional array. 4.2 String- Basic of String, Array of String, Functions in String.h 4.3 Structure- Declaration, Initialization, structure within structure, Operation on structures, Array of Structure.	05	LO4
5	Pointer	5.1 Pointer: Introduction, Definition and uses of Pointers, Address Operator, Pointer Variables, Dereferencing Pointer, Void Pointer, Pointer Arithmetic, Pointers to Pointers, Pointers and Array.	03	LO5
6	Files	6.1 Files: File operation- Opening, Closing, Creating, Reading, Processing File.	02	LO6

Text Books

1. “Basics of Computer Science”, by BehrouzForouzan , Cengage Learning .

2. "Programming Techniques through C", by M. G. Venkateshmurthy, Pearson Publication.
3. "Programming in ANSI C", by E. Balaguruswamy, Tata McGraw-Hill Education.
4. "Programming in C", by Pradeep Day and Manas Gosh, Oxford University Press.
5. "Let Us C", by Yashwant Kanetkar, BPB Publication.

Reference Books

1. "The C Programming Language" by Brian W. Kernighan and Dennis M. Ritchie, Publisher: Prentice Hall
Publication Date: February 22, 1988 ,ISBN-13: 978-0131103627 ,
2. "C Programming: A Modern Approach" by K. N. King, Publisher: W. W. Norton & Company
Publication Date: April 26, 2008 (2nd Edition), ISBN-13: 978-0393979503
3. "C Primer Plus" by Stephen Prata, Publisher: Addison-Wesley Professional Publication Date: December 27, 2013 (6th Edition) ISBN-13: 978-0321928429
4. "Programming in C" by Stephen G. Kochan Publisher: Addison-Wesley Professional
Publication Date: August 18, 2014 (4th Edition) ISBN-13: 978-0321776419

Online Resources:

Sr. No.	Website Name
1.	Learn C - This website offers a free, interactive tutorial to learn C programming, covering both basic and advanced topics.
2.	Codecademy - Codecademy provides a comprehensive, interactive course for learning C, complete with real-world projects and skill paths.
3.	Coursera - Coursera, in collaboration with Duke University, offers a specialization in C programming, including hands-on projects and a certificate upon completion.
4.	edX - This course, offered by edX, covers C programming with a focus on Linux, including professional certification.

Sr No	Suggested List of Experiments	Hrs
01	a) Program to demonstrate Operators Data Input and Output – getchar(), putchar(), scanf(), printf(), gets(), puts() b) Program to demonstrate Operators-Arithmetic, Relational and logical, Assignment, Unary, Conditional, Bitwise, Comma, other operators.	02
02	a) Program to demonstrate Branching - If statement, If-else Statement, Multiway decision. b) Program to demonstrate Looping – while, do-while	02
03	a) Program to demonstrate Nested control structure- Switch statement, Continue statement, Break statement, Goto statement	02
04	a) Program to demonstrate Function, Passing Arguments to a Function (call by value and call by reference	02

05	a) Implement an iterative function for factorial/ Fibonacci etc. b) Implement a recursive function for factorial/ Fibonacci etc.	02
06	a) Program to demonstrate Storage Classes –Auto, Extern, Static, Register	02
07	c) Program to demonstrate Array 1D, d) Program to demonstrate Array 2D	02
08	e) Program to demonstrate String f) Program to demonstrate String arrays of string	02
09	Program to demonstrate Structure Write a program to store and display information of a student/employee etc. using structures. a) Define a structure. b) Read and store details. c) Display the stored information.	02
10	Program to demonstrate pointers a) Define a node structure. b) Implement functions to insert, delete, and display nodes.	02
11	Program to demonstrate files Write a program to maintain a simple student/employee etc. database using file handling. a) Open a file to store student records. b) Implement functions to add, update, and display records. c) Ensure data persistence by saving changes to the file.	02
12	Implement one small application using Function, Files, Structure and Pointers concepts you have learnt in C (eg. : Simple Library Management System 1.Functions: Add, display, and search books. 2. Files: Store and retrieve book data. 3. Structures: Represent a book. 4. Pointers: Manage the list of books dynamically	02

Sr No	List of Assignments / Tutorials	Hrs
01	Flowcharts for programs	02
02	Functions and Parameter	
03	Control Structures	
04	Functions and Parameter	
05	Arrays, String Structure and Union	
06	Pointer and Files	

Assessment :

Term Work: Term Work shall consist of at least 10 to 12 practicals' based on the above list. Also, Term work Journal must include at least 2 assignments.

Term Work Marks: 25 Marks (Total marks) = 15 Marks (Experiment) + 5 Marks (Assignments) + 5 Marks (Attendance)

Practical & Oral Exam: An Oral & Practical exam will be held based on the above syllabus.